

GRAFTON AP, Australia

WMO: 955700

Lat: 29.7583S

Lon: 153.0297E

Elev: 26

StdP: 101.02

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.1	3.5	-3.0	2.9	17.3	-0.8	3.5	13.9	7.2	18.6	6.4	18.6	0.6	190	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	33.8	22.5	31.9	22.2	30.3	21.7	24.5	30.3	23.7	28.9	23.1	28.0	4.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.7	26.1	22.2	17.0	25.6	21.6	16.4	25.1	74.2	30.6	71.2	29.1	68.6	28.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.5	5.9	DB	-0.7	38.9	0.8	2.8	-1.3	40.9	-1.8	42.5	-2.2	44.1	-2.9	46.1	
			WB	-1.1	26.3	0.8	1.1	-1.7	27.1	-2.1	27.8	-2.6	28.4	-3.2	29.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	24.3	23.9	22.6	19.5	16.1	13.9	12.9	13.9	16.9	19.4	21.8	23.0
	DBStd	4.59	2.17	2.16	1.78	1.95	2.22	2.30	2.33	2.34	2.44	2.61	2.46	2.24
	HDD10.0	4	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	604	0	0	0	9	75	134	171	138	56	18	2	1
	CDD10.0	3284	442	390	391	287	189	117	91	123	207	291	353	404
	CDD18.3	842	184	157	132	46	6	1	1	2	13	51	105	146
	CDH23.3	6013	1408	1107	734	236	49	2	7	54	189	459	752	1014
	CDH26.7	1823	501	366	166	26	2	1	0	11	44	141	248	318
Wind	WSAvg	2.4	2.7	2.9	2.5	2.3	1.7	1.8	1.7	2.0	2.3	2.6	2.8	2.8
Precipitation	PrecAvg	1255	150	173	165	98	98	104	50	58	48	76	119	117
	PrecMax	1884	497	436	512	283	450	319	186	232	108	378	221	322
	PrecMin	765	18	32	38	12	1	23	0	0	5	17	20	46
	PrecStd	286	106	107	99	61	101	85	41	65	29	70	53	57
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	35.9	33.1	29.0	26.7	23.1	24.2	27.7	30.8	33.4	35.3	35.6
		MCWB	23.4	23.8	23.0	20.1	17.9	15.8	15.3	15.5	17.2	19.5	20.4	22.4
	2%	DB	33.1	32.6	29.9	27.2	24.6	21.5	22.2	24.6	27.4	30.3	31.7	31.9
		MCWB	23.3	23.7	22.2	19.7	17.3	15.1	14.1	14.7	16.8	19.3	21.4	21.9
	5%	DB	31.1	30.5	28.4	25.9	23.2	20.5	20.6	22.6	25.4	27.8	29.2	29.8
		MCWB	22.8	22.5	21.6	19.0	16.7	14.6	13.7	13.9	16.4	18.5	20.7	21.4
	10%	DB	29.3	28.8	27.0	24.6	21.9	19.4	19.2	20.8	23.5	25.7	27.4	28.2
		MCWB	22.2	21.6	20.8	18.6	15.7	14.1	12.9	13.1	15.7	18.0	20.1	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.9	24.7	22.0	19.8	18.0	18.3	18.1	20.1	22.2	24.1	24.4
		MCDB	32.7	31.4	28.8	25.8	23.8	20.1	21.4	23.9	26.0	28.1	31.5	31.1
	2%	WB	24.2	24.6	23.4	20.9	18.8	16.8	16.5	16.5	18.8	20.8	22.8	23.4
		MCDB	30.2	30.0	27.3	24.7	22.5	19.5	19.2	21.1	24.1	27.2	28.9	29.2
	5%	WB	23.6	23.7	22.6	20.2	17.9	16.1	15.2	15.4	17.8	19.7	21.8	22.7
		MCDB	28.8	28.5	26.4	24.1	21.3	18.6	18.5	19.8	22.7	25.2	26.9	27.8
	10%	WB	23.0	22.7	21.9	19.4	16.9	15.2	14.1	14.4	16.9	19.0	21.0	22.0
		MCDB	27.7	27.2	25.5	23.3	20.3	17.8	17.4	18.9	21.3	23.9	25.2	26.4
Mean Daily Temperature Range		MDBR	10.2	9.7	9.4	10.5	12.6	11.8	14.0	15.3	14.0	12.7	11.3	10.4
	5% DB	MCDBR	13.5	13.1	11.8	12.4	14.0	13.1	15.8	18.1	17.2	17.1	14.9	13.8
		MCWBR	4.9	4.6	4.8	5.5	7.1	7.2	8.3	9.0	7.8	7.2	5.9	5.3
	5% WB	MCDBR	11.2	11.0	9.9	10.4	11.2	9.2	11.5	14.0	14.0	13.7	12.1	11.1
		MCWBR	4.6	4.7	4.7	5.1	6.1	5.4	6.7	7.6	7.1	6.1	5.3	4.7
Clear-Sky Solar Irradiance	taub	0.387	0.373	0.356	0.337	0.311	0.304	0.298	0.312	0.356	0.391	0.386	0.390	
	taud	2.507	2.560	2.604	2.624	2.656	2.661	2.662	2.600	2.471	2.406	2.473	2.486	
	Ebn at Noon	955	950	935	904	885	869	892	918	918	921	949	954	
	Edh at Noon	114	105	95	84	73	69	72	84	106	121	117	117	
All-Sky Solar Radiation	RadAvg	6.51	5.89	4.97	4.24	3.46	2.88	3.31	4.21	5.29	5.90	6.31	6.49	
	RadStd	0.53	0.48	0.32	0.22	0.23	0.18	0.26	0.30	0.38	0.53	0.58	0.60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	-56	+167	+90

Nomenclature: See separate page